

=== CONFIGURACION GNB (1779105932) ===

```
cu_cp:
  amf:
    addr: 10.53.1.2
    port: 38412
    bind_addr: 10.53.1.3
    supported_tracking_areas:
      - tac: 7
        plmn_list:
          - plmn: "20893"
            tai_slice_support_list:
              - sst: 1
    inactivity_timer: 7200

ru_sdr:
  device_driver: zmq
  device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6
  srate: 11.52
  tx_gain: 75
  rx_gain: 75

cell_cfg:
  dl_arfcn: 368500
  band: 3
  channel_bandwidth_MHz: 10
  common_scs: 15
  plmn: "20893"
  tac: 7
  pdcch:
    common:
      ss0_index: 0
      coreset0_index: 12
    dedicated:
      ss2_type: common
      dci_format_0_1_and_1_1: false
  prach:
    prach_config_index: 1
  pdsch:
    mcs_table: qam64
  pusch:
    mcs_table: qam64

log:
  filename: /tmp/gnb.log
  all_level: info
  hex_max_size: 0

pcap:
  mac_enable: false
  mac_filename: /tmp/gnb_mac.pcap
  ngap_enable: false
  ngap_filename: /tmp/gnb_ngap.pcap
  e2ap_enable: true
  e2ap_du_filename: /tmp/gnb_du_e2ap.pcap
  e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap
  e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap

metrics:
  layers:
    enable_rlc: true
    enable_sched: true
  periodicity:
    du_report_period: 1000
```

=== CONFIGURACION UE (1779105932) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 11.52e6
nof_antennas = 1
device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=11.52e6
```

```
[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0
```

```
[rat.nr]
bands = 3
nof_carriers = 1
```

```
[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap
```

```
[log]
all_level = warning
phy_lib_level = none
all_hex_limit = 32
filename = stdout
file_max_size = -1
```

```
[usim]
mode = soft
algo = milenage
opc = 63BFA50EE6523365FF14C1F45F88737D
k = 00112233445566778899aabbccddeeff
imsi = 208930000000001
imei = 353490069873319
```

```
[rrc]
release = 15
ue_category = 4
```

```
[nas]
apn = srsapn
apn_protocol = ipv4
```

```
[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0
```

```
[gui]
enable = false
```

services:

```
5gc:
  container_name: open5gs_5gc
  build:
    context: open5gs
    target: open5gs
    args:
      OS_VERSION: "22.04"
      OPEN5GS_VERSION: "v2.7.0"
  env_file:
    - ${OPEN_5GS_ENV_FILE:-open5gs/open5gs.env}
  privileged: true
  ports:
    - "9898:9999/tcp"
  command: 5gc -c /etc/open5gs/open5gs-5gc.yml
  healthcheck:
    test: [ "CMD-SHELL", "nc -z 127.0.0.5 7777" ] # Check AMF SBI port
    interval: 3s
    timeout: 1s
    retries: 60
  networks:
    ran:
      ipv4_address: ${OPEN5GS_IP:-10.53.1.2}
  configs:
    - open5gs_5gc_config.yml:/etc/open5gs/open5gs-5gc.yml
```

```
gnb:
  container_name: srsran_gnb
  image: srsran/gnb
  build:
    context: ..
    dockerfile: docker/Dockerfile
    args:
      OS_VERSION: "24.04"
  privileged: true
  cap_add:
    - SYS_NICE
    - CAP_SYS_PTRACE
  volumes:
    - /dev/bus/usb/:/dev/bus/usb/
    - /usr/share/uhd/images:/usr/share/uhd/images
    - gnb-storage:/tmp
  configs:
    - gnb_config.yml:/gnb_config.yml
    - gnb_compose_config.yml:/gnb_compose_config.yml
  networks:
    ran:
      ipv4_address: ${GNB_IP:-10.53.1.3}
    metrics:
      ipv4_address: 172.19.1.3
  depends_on:
    5gc:
      condition: service_healthy
  command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml
```

```
mongodb:
  container_name: mongodb
  image: mongo:4.4
  networks:
    ran:
      ipv4_address: 10.53.1.100
  volumes:
    - mongodb-data:/data/db
  healthcheck:
```

```
test: echo 'db.runCommand("ping").ok' | mongo --quiet
interval: 10s
timeout: 5s
retries: 5
restart: always
```

configs:

gnb_config.yml:

```
content: |
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      addr: 10.53.1.2
      port: 38412
      bind_addr: 10.53.1.3
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        - tac: 7
          plmn_list:
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device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6
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tx_gain: 75
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cell_cfg:

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dl_arfcn: 368500
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  common:
    ss0_index: 0
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    ss2_type: common
    dci_format_0_1_and_1_1: false
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log:

```
filename: /tmp/gnb.log
all_level: info
hex_max_size: 0
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pcap:

```
mac_enable: false
mac_filename: /tmp/gnb_mac.pcap
ngap_enable: false
ngap_filename: /tmp/gnb_ngap.pcap
e2ap_enable: true
e2ap_du_filename: /tmp/gnb_du_e2ap.pcap
e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap
e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap
```

metrics:

```
layers:
  enable_rlc: true
```

```
    enable_sched: true
    periodicity:
      du_report_period: 1000
gnb_compose_config.yml:
content: |
  cu_cp:
    amf:
      addr: ${OPEN5GS_IP:-10.53.1.2}
      bind_addr: ${GNB_IP:-10.53.1.3}
  metrics:
    autostart_stdout_metrics: true
    enable_json: true
  remote_control:
    bind_addr: 0.0.0.0
    enabled: true
open5gs_5gc_config.yml:
content: |
  db_uri: mongodb://mongodb/open5gs

logger:
  file: /var/log/open5gs/open5gs.log
  level: info

test:
  serving:
    - plmn_id:
        mcc: 208
        mnc: 93

amf:
  sbi:
    server:
      - address: 127.0.0.5
        port: 7777
    client:
      scp:
        - uri: http://127.0.0.200:7777
  ngap:
    server:
      - address: 10.53.1.2
  metrics:
    server:
      - address: 127.0.0.5
        port: 9090
  guami:
    - plmn_id:
        mcc: 208
        mnc: 93
      amf_id:
        region: 2
        set: 1
  tai:
    - plmn_id:
        mcc: 208
        mnc: 93
      tac: 7
  plmn_support:
    - plmn_id:
        mcc: 208
        mnc: 93
      s_nssai:
        - sst: 1
  security:
    integrity_order : [ NIA2, NIA1, NIA0 ]
    ciphering_order : [ NEA0, NEA1, NEA2 ]
  network_name:
    full: Open5GS
  amf_name: open5gs-amf0
```

```
time:
  t3512:
    value: 540

smf:
  sbi:
    server:
      - address: 127.0.0.4
        port: 7777
    client:
      scp:
        - uri: http://127.0.0.200:7777
  pfcp:
    server:
      - address: 127.0.0.4
    client:
      upf:
        - address: 127.0.0.7
  gtpc:
    server:
      - address: 127.0.0.4
  gtpu:
    server:
      - address: 127.0.0.4
  metrics:
    server:
      - address: 127.0.0.4
        port: 9090
  session:
    - subnet: 10.45.0.0/16
      gateway: 10.45.0.1
    - subnet: 2001:db8:cafe::/48
      gateway: 2001:db8:cafe::1
  dns:
    - 8.8.8.8
    - 8.8.4.4
    - 2001:4860:4860::8888
    - 2001:4860:4860::8844
  mtu: 1400

upf:
  pfcp:
    server:
      - address: 127.0.0.7
  gtpu:
    server:
      - address: 127.0.0.7
  session:
    - subnet: 10.45.0.0/16
      gateway: 10.45.0.1
    - subnet: 2001:db8:cafe::/48
      gateway: 2001:db8:cafe::1
  metrics:
    server:
      - address: 127.0.0.7
        port: 9090

ausf:
  sbi:
    server:
      - address: 127.0.0.11
        port: 7777
    client:
      scp:
        - uri: http://127.0.0.200:7777

udm:
  sbi:
```

```
server:
  - address: 127.0.0.12
    port: 7777
client:
  scp:
    - uri: http://127.0.0.200:7777
```

```
udr:
  sbi:
    server:
      - address: 127.0.0.20
        port: 7777
    client:
      scp:
        - uri: http://127.0.0.200:7777
```

```
nrf:
  sbi:
    server:
      - address: 127.0.0.10
        port: 7777
```

```
pcf:
  sbi:
    server:
      - address: 127.0.0.13
        port: 7777
    client:
      scp:
        - uri: http://127.0.0.200:7777
```

```
metrics:
  server:
    - address: 127.0.0.13
      port: 9090
```

```
nssf:
  sbi:
    server:
      - address: 127.0.0.14
        port: 7777
    client:
      scp:
        - uri: http://127.0.0.200:7777
      nsi:
        - uri: http://127.0.0.10:7777
        s_nssai:
          sst: 1
```

```
bsf:
  sbi:
    server:
      - address: 127.0.0.15
        port: 7777
    client:
      scp:
        - uri: http://127.0.0.200:7777
```

```
scp:
  sbi:
    server:
      - address: 127.0.0.200
        port: 7777
    client:
      nrf:
        - uri: http://127.0.0.10:7777
```

```
volumes:
  gnb-storage:
```

mongodb-data:

networks:

 ran:

 ipam:

 driver: default

 config:

 - subnet: 10.53.1.0/24

metrics:

 ipam:

 driver: default

 config:

 - subnet: 172.19.1.0/24